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The Editorial Board
Astronomy & Astrophysics
EDP Sciences

Dear Editor,

I submit for consideration as a Regular Paper in *Astronomy & Astrophysics*:

**“GAPC: A Catalog of HG Phase Curve Parameters for 128,885
Main-Belt Asteroids from Gaia DR3 Sparse Photometry”**

GAPC is an open-source, reproducible catalog derived from Gaia DR3 sparse photometry, cross-matched with ten external datasets (NEOWISE, LCDB, Bus-DeMeo/PDS taxonomy, Pravec binaries, SDSS MOC4, DAMIT, Goffin masses, GASP spectra, Ast-Dys proper elements, MPC). The catalog contains 128,885 objects and 162 columns; an independent verification script confirms 75/75 checks pass on the released v8.

Main scientific results:

1. **Composition-independent size law.** The partial correlation $r(G, \log D \mid \log p_V) \approx -0.28$ is consistent across all taxonomy complexes (S, M, E, P, C) and is reproduced within asteroid families (Flora $n = 8,168$; Koronis $n = 1,039$; Eunomia $n = 4,963$), pointing to regolith depth as the physical driver rather than mineralogy.
2. **Binary G -excess.** Known binary asteroids (Pravec catalog, $n = 384$) show a statistically significant G -excess after size control ($p = 2.2 \times 10^{-11}$), providing a new observational constraint for binary formation and tidal evolution models.
3. **Phase curves and spectra are orthogonal observables.** Gaia 16-band reflectance spectra achieve $R^2 \approx 0$ in predicting G beyond taxonomy and size, confirming that phase curve shape encodes sub-visual surface texture (grain size, porosity) not accessible through reflectance spectroscopy.

Several physically motivated correlations — family age, rotation period, NEATM thermal inertia, and C-complex hydration — are found to be non-significant and are reported explicitly to avoid publication bias.

The catalog is available on Zenodo (DOI: 10.5281/zenodo.19858420) under MIT license. The full 54-step pipeline is included for reproducibility.

A companion paper (GASP, Scheibenpflug 2026, *Research Notes of the AAS*, vol. 10, p. 87, DOI: 10.3847/2515-5172/ae5e45) describing the spectral catalog used in result (3) has been published.

I declare no conflicts of interest. This manuscript has not been submitted elsewhere and is not under consideration at any other journal.

Sincerely,

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